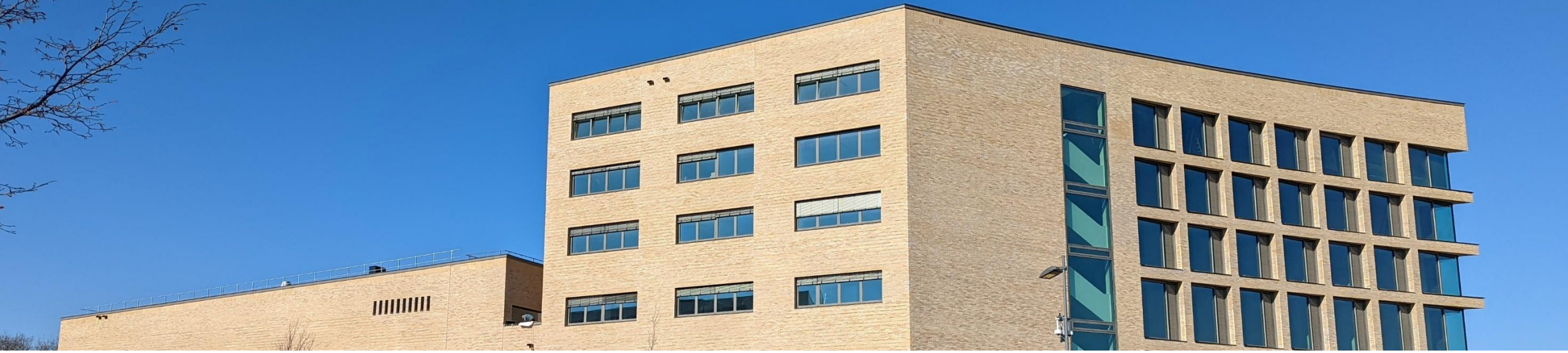


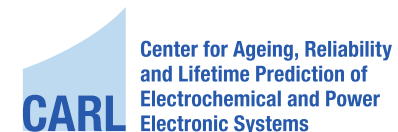


Data-Driven Battery Quality Classification Using Formation Data

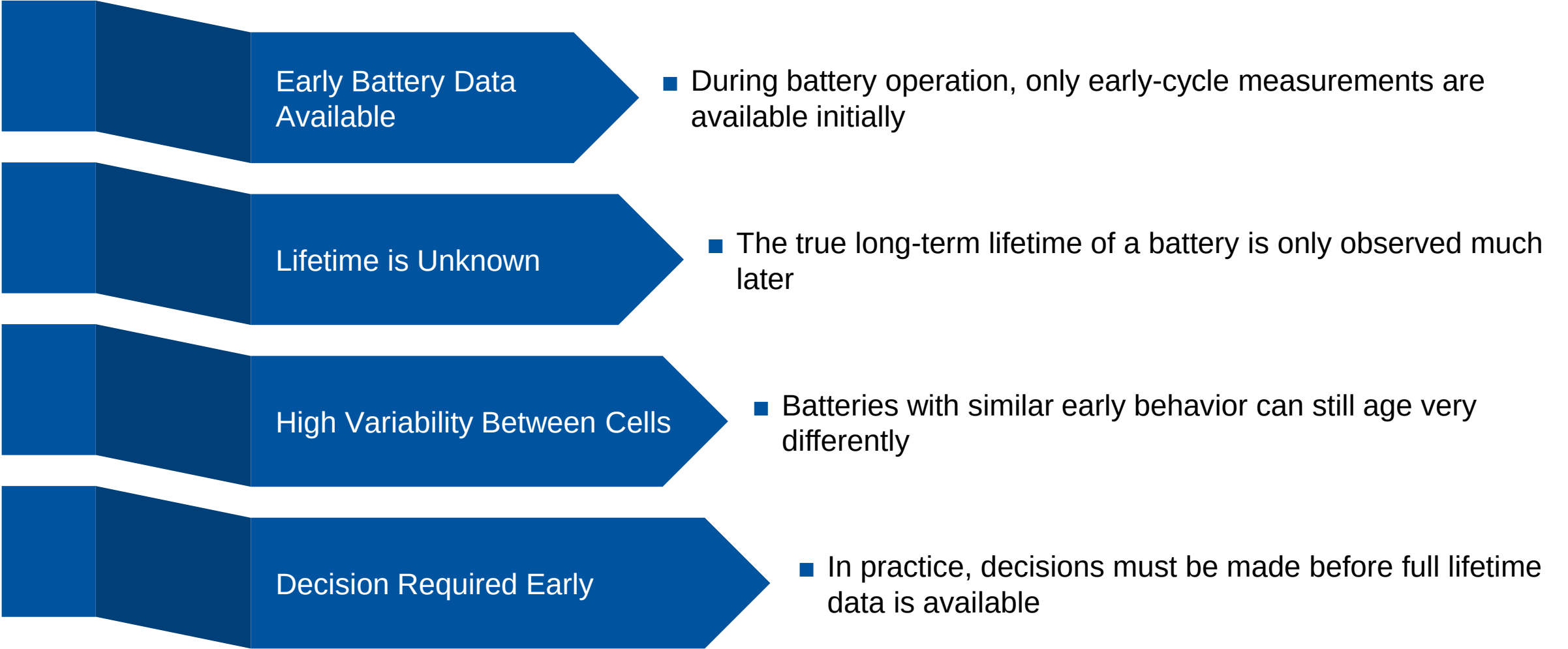
Introduction

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Problem Setup & Motivation



Early Battery Data Available

- During battery operation, only early-cycle measurements are available initially

Lifetime is Unknown

- The true long-term lifetime of a battery is only observed much later

High Variability Between Cells

- Batteries with similar early behavior can still age very differently

Decision Required Early

- In practice, decisions must be made before full lifetime data is available

What Is the Objective of This Dataset?

What we have: Formation data only

- Voltage / current / capacity / time during formation
- Same type of measurements for every cell

What we want: Quality grading (A vs B)

- Class A = higher quality
- Class B = lower quality
- Labels are derived from later ageing outcome (given)

Why it matters: Early quality control in production

- Decide quality before long ageing tests
- Reduce time & cost of end-of-line testing

Key Idea

Classify battery quality using formation data (early, before ageing results are known)

Cells and Experimental Setup

- 40 commercial NMC / graphite lithium-ion pouch cells

- Same production batch
- Two formation protocols: baseline formation / fast formation

- Nominal capacity: 2.36 Ah
- Cell chemistry: NMC / graphite

- Cyclic ageing after formation
- Ageing temperatures: 25 °C and 45 °C

- Industrial battery test system

→ **All cells have identical measurement structure, enabling fair comparison**

Dataset Structure

Processed Dataset Organization

- Raw formation data per cell: **Dataset.xlsx**
- Lifetime information: **CycleOfLife.xlsx**
- Unique identifier: cell_id

Data Formats

- Raw data table:
 - Time, step, current, voltage, charge, discharge, energy
- Target table (per cell):
 - Cycle life value
 - **Important:** class label NOT given directly (must be derived)
- Structured for Python-based analysis

What Data Is Available Per Cycle?

■ Cell identifier

- *cell_id*: uniquely identifies each battery cell

■ Time information

- *time_s*: absolute time stamps
- *step_time_s*: step-wise time stamps

■ Process structure

- *cycle*: formation cycle
- *step*: step index

■ Electrical signals

- *current_A*: cell current
- *voltage_V*: cell voltage

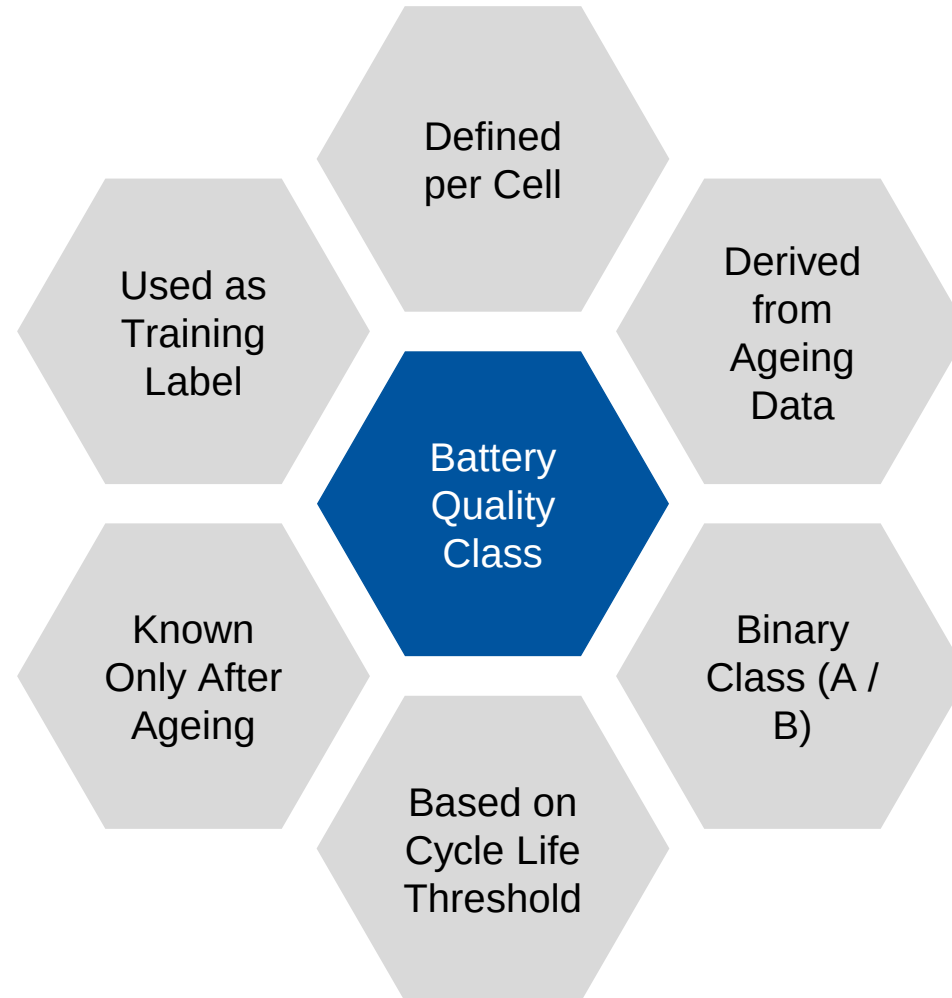
■ Accumulated quantities

- *Q_charge_Ah*: charged capacity
- *Q_discharge_Ah*: discharged capacity
- *E_charge_Wh*: charged energy
- *E_discharge_Wh*: discharged energy

Important notes:

- Multiple rows belong to the same cell and cycle
- A single cell contains hundreds of rows
- You are given raw signals, not features
 - Features must be engineered by aggregating / transforming this data
- Final classification is performed at cell level

Target Variable



Your task: Classify each battery cell as Class A or Class B using only formation data

Key Research Questions

■ Can battery quality be classified reliably using formation data?

■ How do different formation protocols influence battery quality classes?

■ Are simple classification models sufficient to separate battery quality classes?

Feature Engineering: Examples

Time- and Signal-based Features

- Duration spent in specific voltage ranges during formation
- Time evolution of current and voltage during charging steps
- Accumulated charge or energy over formation steps

Formation Process Signatures

- Differences between formation steps or cycles
- Shape characteristics of voltage–time or voltage–charge curves
- Indicators of slow vs fast electrochemical stabilization

Statistical features

- Mean, variance, and range of voltage or current signals
- Signal stability across formation steps
- Noise or fluctuation indicators

Differential and trend-based features

- Differences between consecutive formation steps
- Local slopes or changes in voltage–charge behaviour
- Relative changes in accumulated charge or energy

Project Goal

■ Understand real battery datasets

■ Perform exploratory data analysis

■ Engineer meaningful features

■ Build battery quality classification models

■ Interpret classification results in the context of battery formation physics

Expected Output

Final Project Deliverables

Data Understanding

- Understanding of dataset structure
- Identification of relevant variables
- Basic exploratory data analysis

Feature Engineering

- Features extracted from data
- Justification of selected features
- Physical or process-based interpretation of features

Modeling

- Trained classification model
- Proper train / test split at cell level
- Clear definition of target classes (A / B)

Evaluation & Interpretation

- Classification metrics (e.g. accuracy, confusion matrix)
- Interpretation of misclassifications

Focus on methodology, feature understanding, and decision quality

Literature and Starting Point

■ Mandatory paper:

- Zou et al., Machine learning for battery quality classification and lifetime prediction using formation data, Energy and AI, 2024

■ Additional papers

- Weng et al., J. Power Sources, 2021
- Zhang et al., Energy, 2022
- Li et al., Journal of Energy Storage, 2021